FedECS-LT without a trained generator

A finding from the component ablation, kept out of the paper

The question. The component ablation of FedECS-LT includes one run that the paper does not report: the trained generator of Stage II replaced by class Gaussians built from exact statistics. Everything else is as in FedECS-LT (consensus soft labels, logit adjustment, ten synthetic epochs, the exact readout). Reviewer E asked for this arm: if it matches FedECS-LT, Stage II is unnecessary. It does better than matching.

What the sampler is. After Stage I the clients send, besides the class counts and feature sums, the sum of $hh^\top$ over their features. From these the server forms the class means μ_c and one within-class covariance Σ shared by all classes, shrunk toward a multiple of the identity with $\gamma = 0.1$, the same estimator as the readout's (`fedecs.gaussian_from_sums`). A class-c feature is then $\mathrm{ReLU}(\mu_c + L\varepsilon)$ with $\varepsilon \sim \mathcal{N}(0, I)$ and $LL^\top = \Sigma$ (`models.GaussFeatureGen`). There is no Stage II, no class row, no gate and no moment correction: the sampler has the exact mean and, before the ReLU, the exact pooled covariance by construction. It is the sampler of CCVR, applied to the generator-based pipeline.

The results. Table 1 compares it with the full FedECS-LT on the same splits and seeds (CIFAR-10-LT and CIFAR-100-LT, imbalance ratio 100, ten clients, one per architecture).

Table 1: FedECS-LT with its trained generator and with the Gaussian sampler: balanced accuracy (%) of the heads as trained and with the exact readout, mean ± standard deviation over the seeds that finished; the differences are the sampler minus the full method, paired over common seeds, with uncorrected p; Holm: corrected over the seven component changes of the ablation.

	CIFAR-10-LT	CIFAR-100-LT
Seeds	three	two
FedECS-LT, heads as trained	42.8 ± 2.0	14.3 ± 0.6
Gaussian sampler, heads as trained	45.8 ± 1.3	14.6 ± 0.1
difference (p)	+3.0 (0.022)	+0.5 (0.33)
FedECS-LT with the readout	52.6 ± 1.3	18.7 ± 0.2
Gaussian sampler with the readout	54.3 ± 1.2	19.1 ± 0.1
difference (p; Holm)	+1.7 (0.01; 0.041)	+0.5 (0.31; 0.49)
tail recall difference (p)	+2.1 (0.004)	+0.7 (0.2)
Communication per client (MB)	700 vs 11,278	1,870 vs 12,848
Wall-clock time of a run (s)	2,342 vs 7,910	3,369 vs 4,612

On CIFAR-10-LT the sampler is +1.7 points ahead with the readout, significant after Holm's correction over the seven component changes ($p = 0.041$) and above the noise floor of 0.7 points, and +3.0 points ahead with heads as trained. Its tail recall with the readout is +2.1 points higher. On CIFAR-100-LT it is +0.5 points ahead, from two seeds, which is unresolved. It needs 16.1 times less communication and 3.4 times less time on CIFAR-10-LT (6.9 and 1.4 on CIFAR-100-LT), because Stage II, two hundred rounds of FedAvg on the generator, is the largest part of FedECS-LT's cost.

The same holds for GeFL-F. The paper's variants table already contains GeFL-F with its generator replaced by the same Gaussians, with logit adjustment. That change raises GeFL-F

by +10.3 points as trained on CIFAR-10-LT ($p = 0.001$), its tail recall by +20.6, and its accuracy with the readout by +2.5 ($p = 0.02$).

Why it may work. The paper's third diagnosis is that GeFL-F's generated features do not carry their label: the referee recognises them near chance ($12.9 \pm 2.4\,\%$ on CIFAR-10-LT, where chance is 10%). A Gaussian centred on the exact class mean carries its label by construction, as far as the means differ, and its spread is the real one. The heads then learn the classes a client lacks from features that look like them on average, which is what the generator was meant to provide. That a two-moment sampler beats a trained conditional VAE suggests that, on this shallow frozen feature space, the generator adds nothing that the first two moments do not already carry, and that its training loses part of what they carry. This is an interpretation; the runs do not test it.

What it would mean for the paper. If the sampler became the method, FedECS-LT would be simpler and cheaper: exact class statistics, a Gaussian sampler, consensus soft labels, logit adjustment and the exact readout. The diagnosis would stay as it is; the repair of the generator (anchor, gate, CCWA, moment correction) would be replaced by not training a generator at all, which fits the method's name better. The anchored generator would become an ablation.

What it would cost to adopt. New FedECS-LT runs with the sampler on every dataset where the method is reported: MNIST, Fashion-MNIST and SVHN; the six naturally imbalanced datasets; CIFAR-100-LT seed 5; the new splits (seeds 6–8); and the 50-client setting. Without Stage II a CIFAR run took 2,342 s (CIFAR-10-LT) and 3,369 s (CIFAR-100-LT) on a shared Kaggle GPU. The baselines need no new runs. The method section, the overview figure, Algorithm 1, the cost table and the abstract would change.

Caveats.

- CIFAR-100-LT has two seeds, and the difference there is not resolved.
- Only the two CIFAR benchmarks were run; the gate chose class rows on FER-2013, where the class means are close, and there a two-moment sampler may do worse.
- The sampler releases one more statistic: the $d \times d$ sum of $hh^\top$, $2560^2 \approx 6.6$ million numbers per client on CIFAR. It is shared by all classes, so it reveals less per class than per-class covariances would, but it is second-order information about every client's features. In these runs that sum was computed without secure aggregation (`server.py`, `ssum(..., secure=False)`); a deployment would need it under secure aggregation, at that size.
- The ReLU after sampling makes the generated spread smaller than the exact one, as for the moment correction.